

SKILL-7

1. Process Synchronization and Child Process Monitoring Using waitpid().

To Retrieve PATH Variable, Parse Search Directories, Locate Executables, Verify Permissions, Handle Missing Commands, Test Command Resolution.

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rishitha@Rishitha: ~/skill7
rishitha@Rishitha:~$ mkdir -p skill7
rishitha@Rishitha:~$ cd skill7
rishitha@Rishitha:~/skill7$ pwd
/home/rishitha/skill7
rishitha@Rishitha:~/skill7$ nano waitpid.c
rishitha@Rishitha:~/skill7$ gcc waitpid.c -o waitpid
rishitha@Rishitha:~/skill7$ ./waitpid
Parent Process Started
Parent PID: 620

Parent is waiting for Child...

Child Process Started
Child PID: 621
Child is performing a task...
Child Process Completed

Child PID 621 terminated normally
Child Exit Status: 0
Parent Process Resumed
Parent Process Completed
rishitha@Rishitha:~/skill7$ nano path_resolver.c
rishitha@Rishitha:~/skill7$ gcc path_resolver.c -o path_resolver
rishitha@Rishitha:~/skill7$ ./path_resolver ls
Command found: /usr/bin/ls
rishitha@Rishitha:~/skill7$ ./path_resolver gcc
Command found: /usr/bin/gcc
rishitha@Rishitha:~/skill7$ ./path_resolver abcxyz
Command not found: abcxyz
rishitha@Rishitha:~/skill7$ ./path_resolver pwd
Command found: /usr/bin/pwd
rishitha@Rishitha:~/skill7$ ./path_resolver echo
Command found: /usr/bin/echo
rishitha@Rishitha:~/skill7$ ./path_resolver cat
Command found: /usr/bin/cat
rishitha@Rishitha:~/skill7$ ls
path_resolver path_resolver.c waitpid waitpid.c
rishitha@Rishitha:~/skill7$ |
```